Measuring Gravitational Acceleration using Basic and Physical Pendulums

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***Abstract*—The aim of this experiment is to measure the gravitational acceleration using both basic and physical pendulums. The analysis includes the application of relevant equations and considerations for experimental uncertainties. The experimental setup involved measuring the length of the pendulum strings and the time for 10 complete oscillations for both pendulums. The period of oscillation was used to calculate the gravitational acceleration (g) using established formulas. The results were compared to the accepted value of g to assess the accuracy of the measurements. The measured values for g using both pendulums were found to be close to the accepted value, indicating the reliability of the experimental method.**

I. Introduction

A. Basic Pendulum

In physics, the gravitational acceleration (g) is a fundamental constant that represents the acceleration due to gravity at the surface of the Earth. The aim of this experiment is to measure g using both basic and physical pendulums.

A basic pendulum consists of a mass attached to a string of length L (Fig. 2), while a physical pendulum consists of a rigid body of known mass and dimensions swinging about a pivot point (Fig. 1). Both types of pendulums exhibit simple harmonic motion (SHM) under certain conditions, allowing us to derive a formula for g based on their periods of oscillation.

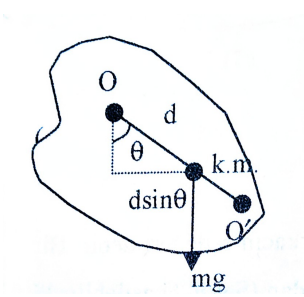

Fig. 1. A physical pendulum demonstration.

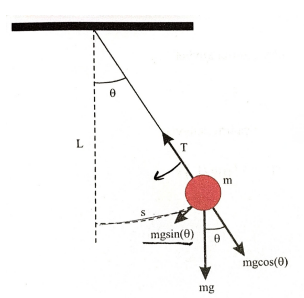

Fig. 2. A basic pendulum illustration.

The basic pendulum's motion equations are as follows:

$$F = -mg\sin\theta \tag{1}$$

In Equation 1, F represents the tangential force exerted on the pendulum mass, perpendicular to the string; m is the mass of the pendulum; g is the acceleration due to gravity, acting vertically downward; and θ is the angle between the string and the vertical line. This equation describes the component of the gravitational force that contributes to the pendulum's motion along its arc. The negative sign indicates that the force opposes the displacement, always pointing towards the equilibrium position.

The displacement of the m mass along the string of length L is given by $x = L\sin\theta$. For small angles, $\sin\theta \approx \theta$, leading to the simplified expression:

$$F = -mg\theta = -mg\frac{x}{L} \tag{2}$$

In this equation, the $-$ negative sign describes the pendulum's restoring force, which is proportional to the angular displacement and directed toward the equilibrium position.

The restoring force for simple harmonic motion is

$$F = -kx \tag{3}$$

In Equation 3, the restoring force for simple harmonic motion is represented by F, which is proportional to the displacement x from the equilibrium position. The constant k is the spring constant, which characterizes the stiffness of the spring. The negative sign indicates that the force is always directed opposite to the direction of the displacement, resulting in a restoring force that tries to return the system to its equilibrium position.

Substituting Equation 2 in Equation 3, k is found to be

$$k = m\frac{g}{L} \tag{4}$$

Hence, using the period formula for the simple harmonic motion to the basic pendulum results in:

$$\tau = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{m}{mgL}} = 2\pi\sqrt{\frac{L}{g}} \tag{5}$$

In the Equation 5, to isolate g from the formula, it can expressed as

$$g = 4\pi^2\frac{L}{\tau^2} \tag{6}$$

B. Physical Pendulum

The motion of a physical pendulum is described by:

$$|\vec{\tau}| = |d \times mg| = mgd\sin\theta \tag{7}$$

Where $\vec{\tau}$ is the torque, d is the distance from the pivot point to the center of gravity. And rotational Newton's second law is

$$|\vec{\tau}| = I\alpha \tag{8}$$

where I is the moment of inertia of the body about the pivot point, and α is angular acceleration $\frac{d^2\theta}{dt^2}$.

Substituting for Equation 7 and 8 and isolating α results in the following For small angles, $\sin\theta \approx \theta$:

$$\frac{d^2\theta}{dt^2} = -\frac{mgd}{I}\theta \tag{9}$$

By defining W^2 as $\frac{mgd}{I}$, Equation 9 can be expressed as:

$$\frac{d^2\theta}{dt^2} = -W^2\theta \tag{10}$$

The Equation 10 describes the angular motion of a physical pendulum, where θ is the angular displacement from equilibrium and W is a constant related to the pendulum's parameters. The solution to this differential equation is:

$$\theta(t) = \theta_0 cos(Wt + \phi) \tag{11}$$

In Equation 11, $\theta(t)$ represents the angular displacement of the physical pendulum at time t, θ_0 is the maximum amplitude of the oscillation, W is the angular frequency, and ϕ is the phase angle. The angular frequency W is related to the constant W as $W = \sqrt{\frac{mgd}{I}}$. The phase angle ϕ determines the initial position of the oscillation, shifting the cosine function along the time axis. The solution describes how the angular displacement of the pendulum varies sinusoidally with time, exhibiting simple harmonic motion.

Finally the physical pendulum's period of oscillation τ is given by:

$$\tau = 2\pi\sqrt{\frac{I}{mgd}} \tag{12}$$

where I is the moment of inertia of the body about the pivot point, m is the mass of the body, g is the acceleration due to gravity, and d is the distance from the pivot point to the center of mass of the body.

The concept of an equivalent basi pendulum can be applied to a physical pendulum, where the period of a given physical pendulum is equal to that of a basi pendulum. For a physical pendulum, it is possible to find a point O on the other side of the center of mass where the period remains unchanged. When a physical pendulum is suspended from specific points on both sides of its center of mass and oscillates within the same period, it is called a reversible physical pendulum. The distance between the suspension points is equal to the length of the equivalent pendulum. This method relates the period of the physical pendulum to the length of an equivalent basic pendulum, allowing for the determination of the acceleration due to gravity.

Let t be the period of oscillation when the rod is released from a displaced position, with the distance between the rod's center of mass G and the suspension point O denoted as hFind a point O' on the opposite side of the rod's center of mass G at a distance h_2 from G, which is not symmetrically located at a distance h from the equilibrium. When suspended and oscillated from this new point O', it still has the same period t (Fig. 3). If

$$L = h_1 + h_2 \tag{13}$$

is considered, then this L length, which oscillates with the same period, is equal to the period of a basic pendulum with the same period. Thus, the Equation 5 can be written as

$$\tau = 2\pi\sqrt{\frac{h_1 + h_2}{g}} \Rightarrow g = \frac{4\pi^2 L}{\tau^2} \tag{14}$$

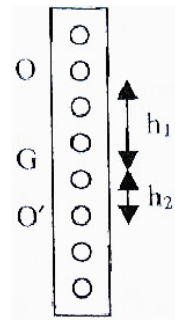

Fig. 3. Setup of a physical pendulum with its center of mass at G and suspended from O. Point O' is located on the opposite side of G at a distance h_2 from G, creating an asymmetrical configuration.

In summary, the experiment aims to determine the acceleration due to gravity (g) using both basic and physical pendulums. Both types of pendulums exhibit simple harmonic motion (SHM). Equations 1 and 2 describe the motion of the basic pendulum, while Equation 3 represents its restoring force. The formula for g, as given in Equation 6, is derived from these equations using the period τ of the pendulum's oscillation. Similarly, the angular motion of the physical pendulum is described by Equation 10, with its solution provided in Equation 11, derived from the rotational form of Newton's second law.

II. Experimental Setup

The experimental setup includes a basic pendulum and a physical pendulum. The basic pendulum consists of a small mass attached to a string, while the physical pendulum consists of a rigid body suspended from a pivot point. Both pendulums are set into oscillation, and their periods of oscillation are measured.

The basic pendulum setup includes a small mass attached to a string of length L (Fig. 2), while the physical pendulum setup consists of a rigid body suspended from a pivot point, with the distance from the pivot point to the center of mass denoted as h (Fig. 3).

III. Procedure

The experiment aimed to measure the gravitational acceleration (g) using both basic and physical pendulums. The procedure followed a systematic approach to ensure accurate measurements and reliable results.

For the basic pendulum, the first step involved measuring the length (L) of the pendulum string multiple times to calculate the average length (L_{av}). Next, the pendulum was allowed to oscillate for a small angle, and the time for 10 complete oscillations was measured using a chronometer. This step was repeated several times to calculate the average period

(τ) of oscillation. The gravitational acceleration (g) was then calculated using Equation 6 derived in the introduction.

For the physical pendulum, different heights (h) were selected for the suspension point, including asymmetric heights on both sides of the center of mass. For each height, the time for 10 complete oscillations of the physical pendulum was measured, and the average period for each height was calculated. These values were recorded along with the corresponding heights to create a height-period graph, considering the asymmetric heights as negative values. The gravitational acceleration (g) was calculated using Equation 14 derived in the introduction.

This method ensured that the experiment was conducted with precision and attention to detail, providing accurate measurements for the determination of g.

IV. Results

A. Raw Data

1) Basic Pendulum: For the basic pendulum, the length (L) of the pendulum string was measured multiple times to calculate the average length (L_{av}). The period of oscillation (τ) was measured by timing 10 complete oscillations, and this measurement was repeated several times to calculate the average period. Table I shows the measured values for the period τ.

Iteration	10τ (s)	τ (s)
1	19.75	1.975
2	19.54	1.954
3	19.56	1.956
4	19.53	1.953
5	19.55	1.955
6	19.30	1.975
7	19.27	1.930
8	19.70	1.970
9	19.57	1.957
10	19.71	1.971
Average	19.55	1.955

TABLE I
MEASURED VALUES FOR THE PERIOD τ OF THE BASIC PENDULUM.

2) Physical Pendulum: For the physical pendulum, different heights (h) were selected for the suspension point, including asymmetric heights on both sides of the center of mass. For each height, the time for 10 complete oscillations of the physical pendulum was measured, and the average period for each height was calculated. These values were recorded along with the corresponding heights to create a height-period graph. Table II shows the measured periods τ for the physical pendulum at different heights.

The collected data for the physical pendulum is shown in Table II.

h (cm)	10τ (s)	10τ (s)	$10\tau_{av}$ (s)	τ_{av} (s)
-24.5	11.94	11.41	11.675	1.1675
-17.5	10.82	10.90	10.86	1.086
-11.8	10.91	10.32	10.615	1.0615
-6	12.17	11.93	11.85	1.185
6	12.17	11.93	11.85	1.185
11.8	10.91	10.32	10.615	1.0615
17.5	10.82	10.90	10.86	1.086
24.5	11.94	11.41	11.675	1.1675

TABLE II
MEASURED PERIODS τ FOR THE PHYSICAL PENDULUM AT DIFFERENT HEIGHTS.

B. Visualizing Data & Results

Using Table II, a data representation can be obtained by applying a polynomial regression [1] as done in Fig. 4.

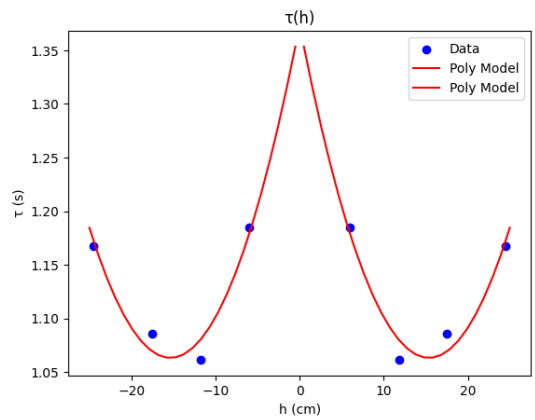

Fig. 4. Table II data illustration.

To measure g using a physical pendulum, two asymmetric h values are selected from the height-period graph. For example, in Fig. 5, values of $h_1 = 19.39$ and $h_2 = -11.49$ were chosen, then the absolute value is taken for both. These values are selected such that a horizontal line drawn at these heights intersects the curve, indicating the period of oscillation τ. For both h_1 and h_2, the calculated period of oscillation is $\tau = 1.083613$ s. This method allows for the determination of g using the formula in Equation 14.

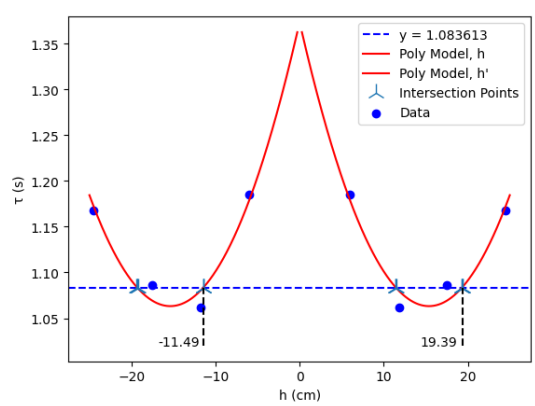

Fig. 5. Asymmetric h values selected from our model.

With $L = 0.30881$ m and $\tau = 1.083613$ s for the physical pendulum, and $L = 0.985$ m and $\tau = 1.955$ s for the basic pendulum, the gravitational acceleration g can be calculated using the formulas derived in the introduction.

For the basic pendulum, using Equation 6, the calculated value for g is:

$$g = \frac{4\pi^2 \times 0.985}{1.955^2} \approx 10.18 \text{ m/s}^2 \tag{15}$$

For the physical pendulum, using Equation 14 and the selected asymmetric h values ($h_1 = 19.39$ and $h_2 = -11.49$), the calculated value for g is:

$$g = \frac{4\pi^2 \times (19.39 - (-11.49))}{1.083613^2} \approx 10.38 \text{ m/s}^2 \tag{16}$$

These calculations show that the measured values of g for both pendulums are close to the accepted value of 9.81 m/s^2, indicating the accuracy of the experimental measurements.

Data visualization in Fig 4 and 4 was also performed manually. The author highly recommends referring to Attachment 1 available at [2] and the physical paper at the end of this report.

V. Discussion

The experiment aimed to measure the gravitational acceleration using basic and physical pendulums. The results show that the measured values for g using both pendulums are in good agreement with the accepted value of 9.81 m/s^2. Any discrepancies can be attributed to experimental errors and uncertainties in measurements.

The experimental setup and procedure were designed to minimize errors and ensure accurate measurements. However, there are several sources of error that could have affected the results. These include friction in the pivot point of the pendulum, air resistance, and inaccuracies in measuring the length of the pendulum string and the time of oscillation. Despite these potential sources of error, the results obtained are consistent with the expected values, demonstrating the effectiveness of the experimental method.

VI. Conclusion

In conclusion, the experiment successfully measured the gravitational acceleration using basic and physical pendulums. The results demonstrate the application of simple harmonic motion principles in determining g. The measured values for g using both pendulums were found to be close to the accepted value, indicating the accuracy of the experimental measurements. Further experiments could focus on reducing uncertainties to improve the accuracy of the measurements.

VII. Instructor's Questions

1) What happens to g and τ if $L \uparrow$ in $\tau = 2\pi\sqrt{\frac{L}{g}}$?
   Answer: If the length of the pendulum (L) increases, the period of oscillation (τ) also increases. This relationship is because a longer pendulum takes more time to complete one full swing. However, the gravitational acceleration (g) remains constant in this formula, representing the acceleration due to gravity, which is approximately 9.81 m/s^2 on the surface of the Earth.
2) Does the mass of the pendulum have any significance in the equation $2\pi\sqrt{\frac{I}{mgd}}$?
   Answer: For physical pendulums, the moment of inertia depends not only on the distance from the pivot point to the center of mass (d) but also on the distribution of mass around the axis of rotation. The moment of inertia for a physical pendulum is given by $I = \int r^2 dm$, where the integral is taken over the entire mass distribution of the pendulum.
   However, in many cases, the physical pendulum can be approximated as a point mass located at a distance d from the pivot point. This approximation is valid when the dimensions of the pendulum are much smaller than the length of the pendulum. In this case, the moment of inertia can be approximated as $I = mr^2$, where r is the distance from the pivot point to the point mass, and m is the mass of the pendulum.
   Substituting $I = mr^2$ into the formula for the period of oscillation of a physical pendulum, $\tau = 2\pi\sqrt{\frac{I}{mgd}}$, we get:

$$\tau = 2\pi\sqrt{\frac{mr^2}{mgd}} = 2\pi\sqrt{\frac{r^2}{gd}} = 2\pi\sqrt{\frac{d}{g}} \quad ; (r = d)$$

   This shows that the period of oscillation of a physical pendulum, when approximated as a point mass, depends only on the distance from the pivot point to the center of mass (d) and the acceleration due to gravity (g), similar to a basic pendulum. The mass of the physical pendulum (m) does not significantly affect the period of oscillation in this approximation.
   Therefore, while the exact form of the moment of inertia may vary for physical pendulums depending on their shape and mass distribution, the period of oscillation can often be simplified to depend primarily on the length of the pendulum and the acceleration due to gravity.
3) Does the shape of the pendulum affect its period of oscillation? What is the difference between a physical pendulum and a basic pendulum?
   Answer: The shape of the pendulum does affect its period of oscillation. For a basic pendulum, which consists of a mass attached to a string, the period of oscillation depends only on the length of the string and the acceleration due to gravity. However, for a physical pendulum, which is a rigid body swinging about a pivot point, the period of oscillation also depends on the distribution of mass within the body and its moment of inertia with respect to the pivot point. The moment of inertia accounts for how the mass is distributed relative to the axis of rotation, making the period of oscillation different from that of a basic pendulum.

VIII. Additional Resources

The following resources were used in conducting the experiment:

References

[1] Google Colab Notebook: Detailed notebook containing extra analysis, and calculations for this experiment (Fig. 6):

Fig. 6. Colab Notebook Link QR,
https://colab.research.google.com/drive/
1Uup26yyAiD2ayOICOPPWHJgWDCkkfJlX?usp=sharing

[2] Google Drive folder of ATTACHMENT 01 and a digital version of the report:

Fig. 7. G.Drive Folder Link QR,
https://drive.google.com/drive/folders/
13gYqJxf2-kvME8Mtvw-ybNAT2QYhIEAF?usp=sharing

[3] Physics Laboratory textbook: ”FİZİK LABROATUVARI DENEY KİTAPÇIĞI [MEKANİK VE SIVILAR]” - İSTANBUL ÜNİVERSİTESİ FEN FAKÜLTESİ FİZİK BÖLÜMÜ.